

Retraction

Retracted: Meta-Analysis on the Effects of Electric Acupuncture on Neural Functional Recovery and Related Pathways of Rats after Spinal Cord Injury

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This article has been retracted by Hindawi, as publisher, following an investigation undertaken by the publisher [1]. This investigation has uncovered evidence of systematic manipulation of the publication and peer-review process. We cannot, therefore, vouch for the reliability or integrity of this article.

Please note that this notice is intended solely to alert readers that the peer-review process of this article has been compromised.

Wiley and Hindawi regret that the usual quality checks did not identify these issues before publication and have since put additional measures in place to safeguard research integrity.

We wish to credit our Research Integrity and Research Publishing teams and anonymous and named external researchers and research integrity experts for contributing to this investigation.

The corresponding author, as the representative of all authors, has been given the opportunity to register their agreement or disagreement to this retraction. We have kept a record of any response received.

References

- [1] Z. Lai, H. Liu, and G. Liu, "Meta-Analysis on the Effects of Electric Acupuncture on Neural Functional Recovery and Related Pathways of Rats after Spinal Cord Injury," *BioMed Research International*, vol. 2022, Article ID 8613384, 12 pages, 2022.

Research Article

Meta-Analysis on the Effects of Electric Acupuncture on Neural Functional Recovery and Related Pathways of Rats after Spinal Cord Injury

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Background. Spinal cord injury (SCI) is a type of damage to the central nervous system (CNS) caused by various factors. The secondary injury of SCI is more complicated. Studies have found that electroacupuncture can help the recovery of nerve function during spinal cord injury. Therefore, this study explored the efficacy of electroacupuncture on complications after spinal cord injury through meta-analysis. **Methods.** Relevant literatures published from January 2010 to March 2022 were searched with “Electric acupuncture, Spinal cord injury, Neural functional recovery, Spinal cord injury in rats, Neuronal Signaling” as search words. The risk of bias of included references was analyzed and assessed using RevMan 5.3 software and Stata software. Heterogeneity between studies was assessed using the *Q*-test and heterogeneity (I^2). **Results.** There was no heterogeneity among the study groups. The comparison on the therapeutic effects of electroacupuncture and conventional therapy suggested that electroacupuncture was more effective for nerve recovery after spinal cord injury than conventional therapy. It can better improve the recovery of motor function after spinal cord injury in rats. On the other hand, SCI+EA had a good inhibitory effect on the expression of RhoA protein signal in rats and had a positive effect on the signal pathway. **Discussion.** The results of meta-analysis confirmed that electroacupuncture was more effective than conventional therapy in inhibiting pathways.

1. Introduction

Spinal cord injury (SCI) is a type of damage to the central nervous system (CNS) caused by various factors, and about 90% of SCI is caused by external trauma. It can cause damage to the continuity and integrity of the spinal cord, and its disability rate is high. In addition, it is often accompanied by complications of various systems, causing both physical and psychological damage to patients [1]. Statistics show that the premature mortality rate of SCI patients is five times higher than that of normal people. In the United States, the lifetime treatment cost of SCI patients is 1.1–4.7 million US dollars, which brings a heavy burden to individuals, families, and

society. Therefore, the mechanism of tissue repair after SCI and the enhancement of damaged neuroplasticity are the current research hotspots and difficulties in neurorehabilitation field [2]. SCI can be divided into two pathways: the primary injury is irreversible due to the trauma itself, and the secondary injury, as the name suggests, is the cascade effect triggered by the primary injury. The secondary injury process is complex, involving more than 20 related mechanisms that affect each other and jointly cause changes in the micro-environment of the SCI area, resulting in difficulties in repairing nerve tissue. However, because of its controllability, early intervention can be used to maximize the repair of damaged nerves. Disrupted axonal reconnection is an

important factor in repairing damaged nerve function [3, 4]. It is currently believed that extracellular inhibitory signaling factors are secreted in large quantities in the damaged area and eventually activate RhoA and its downstream pathways, thereby affecting the cytoskeleton structure of damaged cells and inhibiting the regeneration of nerve growth cones, which in turn leads to limited axon extension.

Electric acupuncture (EA) therapy is to improve the therapeutic effect of traditional acupuncture with the help of electrical stimulation after the acupuncture points “get Qi.” The dual effects of acupuncture and electrical stimulation not only act on the local microenvironment of the Yu poun, but also transmit along the nerves, muscle fibers, and other carriers, causing the relevant muscles to excite, prompting the nerve branches to release nerve skin and neurotransmitters, and promoting neural stem cells. It plays an important role in the treatment of neurological diseases by regulating the related pathways to repair the damage. It has a good effect on the complications of neuropathic pain, neurogenic bladder, and digestive tract disorders after SCI [5–7]. Basic studies have found that EA stimulation of Dazhui and Mingmen acupoints can inhibit the expression of GluR1 in SCI rats, reduce the excitotoxicity of glutamate, and promote the recovery of neuronal and hindlimb motor functions [8]. Relevant studies suggest that electroacupuncture can regulate the functions of the pelvic floor muscles and anal sphincter by stimulating the anterior root of the iliac nerve. Compared with the primary damage of spinal cord injury, the consequences of secondary damage are more serious to a certain extent but can be controlled by early intervention, which has become a key factor in the current treatment of SCI [9–11]. In spinal cord injury, RhoA and its downstream factor ROCK (Rho kinase), as a negative factor signaling pathway for growth cone motility, are blocked by the coreceptor NgR of various inhibitory factors (MAG, CSPGs, semaphorin 4D, etc.) in the local microenvironment and complex p75NTR activated. It mediates the transfer of inflammatory cytokines, causes the collapse and retraction of growth cones, and induces apoptosis of glial cells and neurons, thereby inhibiting axonal elongation. Therefore, inhibition of this signaling pathway is an important target in the treatment of spinal cord injury [12, 13].

In this meta-analysis, it innovatively incorporated current domestic and foreign literature studies on the effects of EA on neural functional recovery and related pathways in rats after SCI. The effect of EA and common methods on neural functional recovery and related pathways in rats with SCI was systematically evaluated and compared by meta-analysis, so as to evaluate the reference value of EA in helping neural functional recovery in SCI, providing theoretical reference for the follow-up treatment for SCI.

2. Materials and Methods

2.1. Literature Retrieval. The PubMed, MEDLINE, EBSCO, Science Direct, Cochrane Library, and China National Knowledge Infrastructure (CNKI) were retrieved to screen related literature. Relevant literatures published from January 2000 to March 2022 were searched using “Electric acu-

puncture, Spinal cord injury, Neural functional recovery, Spinal cord injury in rats, Neuronal Signaling” as search words. Professional journals were manually searched to avoid omissions, and the research subjects were rats.

The retrieval process used a combination of subject headings and free words to conduct multiple searches to obtain references that can be included and then used a search engine to track down each document. The quality of included literature was assessed using RevMan 5.3 software provided by the Cochrane Collaboration.

2.2. How to Include and Exclude the Literature. The literature meeting the following criteria can be included: therapeutic research on neural functional recovery after SCI; histopathological treatment with EA as the gold standard; research subjects which were rats; literature with available true positive, false positive, false negative, and true negative values; and sample size in the study which was greater than 10 cases.

Literatures satisfying the following conditions had to be excluded: literatures such as reports, reviews, conferences, letters, journals, and evaluations; histopathological characteristics which were not used as the gold standard in the study; literatures with less than 10 subjects in the study; and literatures with insufficient data for identifying therapeutic indicators.

2.3. Data Extraction. Two professionals used unified Microsoft Excel (Microsoft, the United States) to independently screen the literature and extract data, cross-check and include the final results, and resolve any differences through discussion. The main extracted data included general information about the included studies: title, first author, and publication year. Basic characteristics of research subjects are as follows: number of rats and experimental methods; the results of the treatment index test; the detection success rate of neural functional recovery after SCI in each study and the data to determine the accuracy of the test (sensitivity and specificity); and the data related to impacts on effect of EA treatment on the signal pathway (evaluation index of whether it affected or not).

2.4. Literature Assessment. The quality of included studies was assessed using the quality assessment of diagnostic accuracy studies (QUADAS) criteria recommended by Cochrane (the United States). The quality of the included original literature was evaluated according to each evaluation index, and each study was evaluated according to “consistent,” “inconsistent,” and “uncertain.”

2.5. Statistical Methods. Risk of bias assessment of the included literatures was performed using RevMan 5.3 software (Cochrane, the United States) and Stata software (Stata Corp, the United States). The heterogeneity among the literatures was assessed using the Q-test and heterogeneity (I^2), and the sensitivity and specificity of common treatment and EA+SCI treatment were calculated and compared, expressed in 95% confidence interval (CI). Forest plots and summary receiver operating characteristic (SROC) curves were drawn, and the funnel plots for asymmetric linear regression were drawn, too. Funnel plots for different treatment measures were used to test for potential publication bias, and sensitivity analyses were performed.

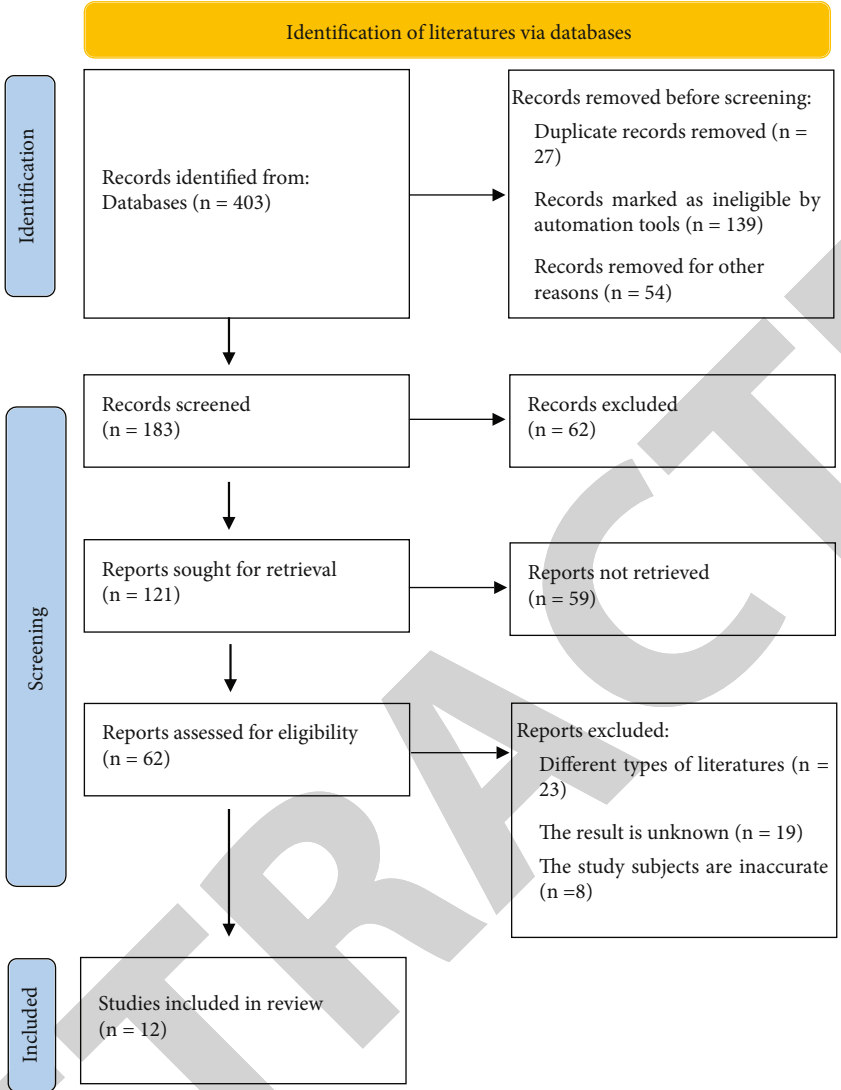


FIGURE 1: Flowchart of searching literature.

TABLE 1: Basic information of the included articles.

Author	Year	Cases	Gender	Weight	Damage section	Frequency	Quality
Li [18]	2018	72	Male	200-250 g	T10	2 Hz-15 min	B
Liang [19]	2014	60	Male	220-250 g	T9-T10	2 Hz-15 min	A
Min [20]	2017	80	Male	220 ± 20 g	T10	2 Hz-20 min	C
Song [21]	2019	18	Male	260-300 g	S1-5	1 Hz-20 min	A
Tu [22]	2018	60	Male	200-250 g	S1-5	2 Hz-30 min	B
Xiao [23]	2019	64	Male	200 ± 20 g	T10	2 Hz-20 min	C
Yan [24]	2011	36	Female	220-250 g	T10	2 Hz-20 min	A
Zhang [25]	2016	80	Male	230-270 g	T10	20 Hz-30 min	A
Zhang [26]	2017	54	Male	250 ± 20 g	T9-T11	2 Hz-30 min	B
Zhao [27]	2020	96	All	—	T8-9	2 Hz-30 min	A
Zheng [28]	2020	48	Male	180-220 g	L5	100 Hz-1 min	A
Zhu [29]	2017	24	Male	180-220 g	T10	2 Hz-30 min	A

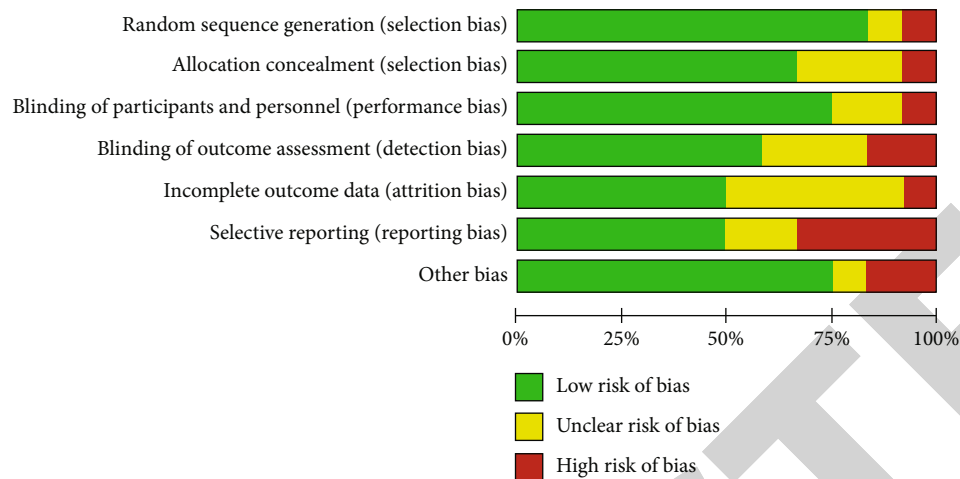


FIGURE 2: Bias risk assessment results of references drawn by RevMan 5.3 software.

3. Results

3.1. Search Results and Basic Information of Literature. A total of 403 articles were obtained by searching databases (323) and manually searching journals (80). First, 27 duplicate publications were excluded, 139 unqualified papers and 54 other reasons were excluded, and the remaining 183 papers were initially selected. By reading abstracts and titles, 62 articles were excluded and 121 articles remained. 59 research reports and review articles were excluded, remaining 62 articles. All the remaining articles were read in full text one by one, and 23 articles with incorrect research types were excluded; 19 articles were excluded if the required treatment results were incomplete or unavailable; and there were 8 articles whose subjects were not rats with spinal cord injury, and 12 articles were finally included in the meta-analysis. Figure 1 shows a flow chart for searching literature.

Basic information of the included literature was extracted by reading the content of the literature. Among the 12 included literatures, 366 were treated with SCI rats + electroacupuncture, and 335 were treated with conventional therapy. In addition, in the 12 included articles, the sample size ranged from 18 to 96. The 12 articles described in detail the various therapeutic indicators of electroacupuncture for spinal cord injury to help the recovery of nerve function and the effect of electroacupuncture on signaling pathways, all of which took the histopathological treatment obtained by surgery or acupuncture as the gold standard. The quality evaluation of the 12 included articles showed that 7 (58.33%) were rated A, 3 (25.00%) were rated B, and 2 (25.00%) were rated C (16.67%). Table 1 lists the basic characteristics of the included literature. Figures 2 and 3 were the reference risk bias assessment graph and reference risk bias summary graph drawn by RevMan 5.3 software, respectively.

3.2. Heterogeneity Assessment Results. The heterogeneity of the two treatment methods in the included articles [14–25] was evaluated, and the results showed that there was no heterogeneity among these articles in terms of neurological recovery ($I^2 = 0.00\%$). There were no heterogeneity among

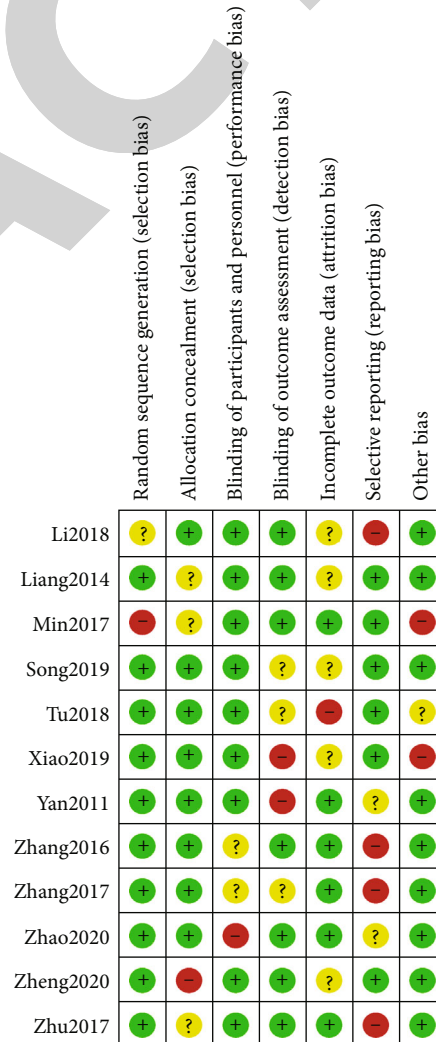


FIGURE 3: Summary graph of reference risk bias. Note: “+” meant low risk, “-” meant high risk, and “?” meant unclear risk.

the included articles in motor function ($I^2 = 0.00\%$) and no heterogeneity among articles in functional recovery ($I^2 = 0.00\%$). In order to further verify the heterogeneity

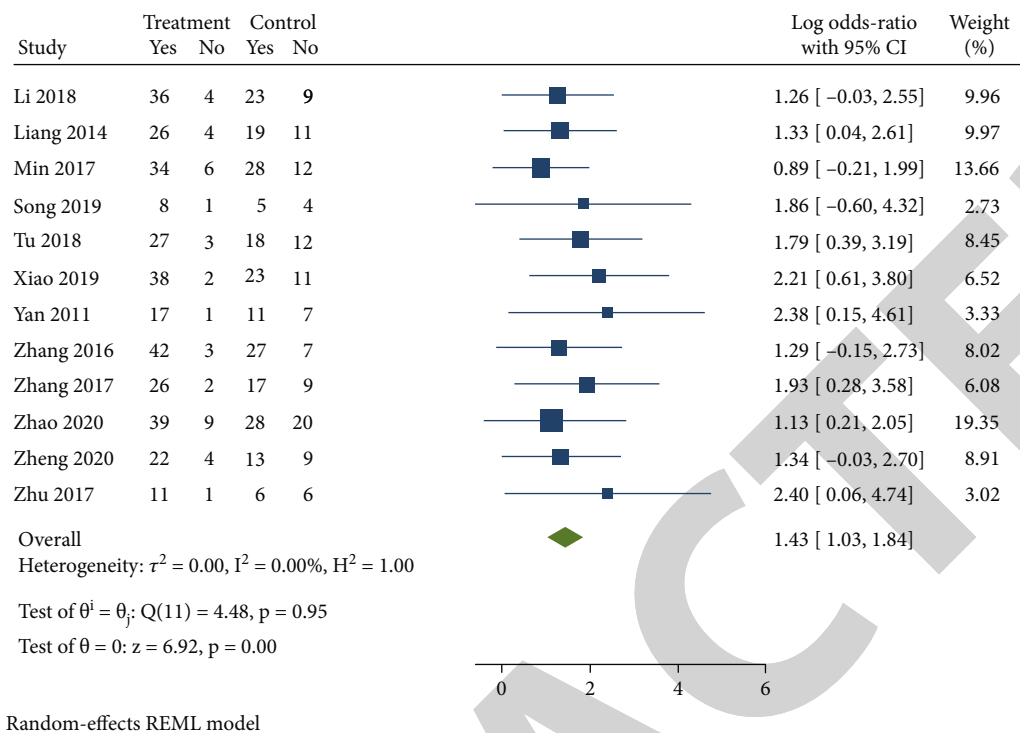


FIGURE 4: Forest plot of individual and pooled studies of neural recovery. CI: confidence interval; df: degree of freedom.

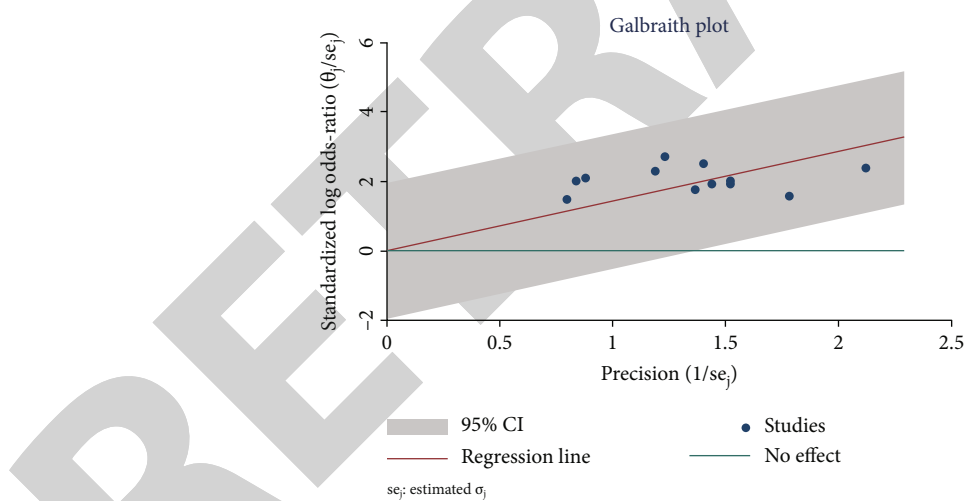


FIGURE 5: Galbraith’s test diagram of the effect of nerve recovery treatment.

between the data of the two treatment methods, a random-effects model should be used for pooled analysis and fitted with a heterogeneity test.

3.3. Therapeutic Effects of Neurological Restoration. In the 12 included articles, the comparison results of conventional treatment and electroacupuncture in the treatment of SCI in the experiment were analyzed. Figure 4 is a forest plot of individual studies and pooled studies for the two treatments. Comparing the effects of electroacupuncture and conventional therapy on nerve recovery in 12 literatures, the overall OR value was 1.43 (1.03, 1.84). The results of het-

erogeneity test were as follows: $Q = 4.48$, $I^2 = 0.00\%$, and $P = 0.95$. There was no heterogeneity among the study groups. Among them, the lowest OR value was 0.89 (-0.21, 1.99), and the highest OR value was 2.40 (0.06, 4.74). The results of this meta-analysis demonstrate that electroacupuncture was more effective than conventional therapy for neurological recovery after SCI.

In order to further observe the effect of the treatment, the heterogeneity test and the risk of bias assessment were carried out for the treatment. As can be seen from Figures 5 and 6, the heterogeneity gap between the studies was small and the accuracy was high and there was a certain

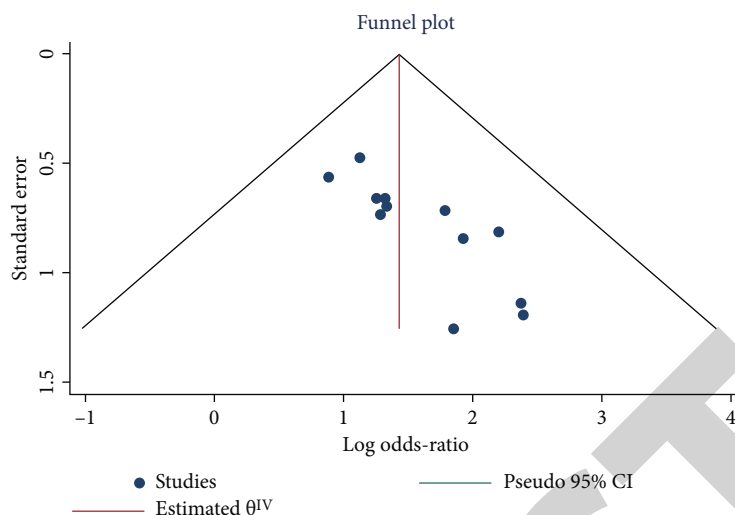


FIGURE 6: Heterogeneity test chart of the effect of nerve recovery treatment.

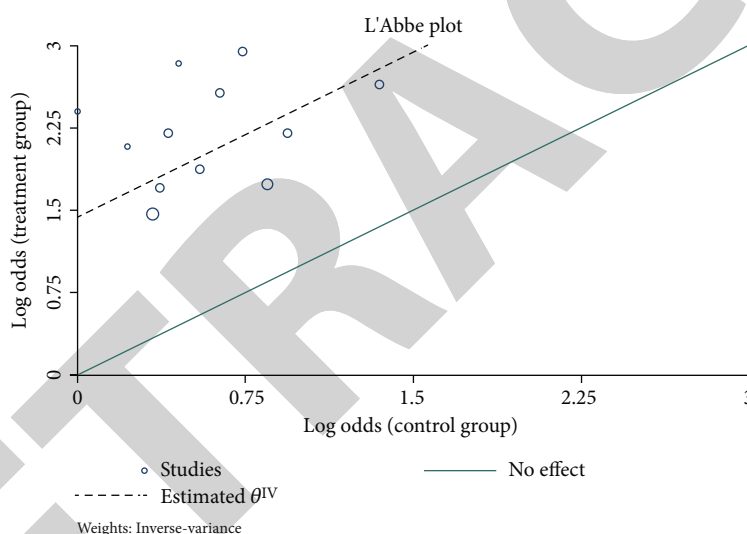


FIGURE 7: Funnel plot of nerve recovery treatment effect.

gap between the two treatment effects. All the literatures had been tested by Western blotting, and the conventional treatment of SCI had not shown a good therapeutic effect on nerve recovery after SCI in rats. Figure 7 shows the funnel plot. The funnel asymmetric linear regression analysis can further observe that there was no risk of bias in the study, and each study was relatively concentrated, which proved that the results had high accuracy.

3.4. Therapeutic Effect of Motor Function. In the 12 included literatures, the comparison results of conventional treatment and electroacupuncture in the treatment of SCI in the experiment were analyzed. Figure 8 is a forest plot of individual studies and pooled studies for two treatments. The effects of electroacupuncture and conventional therapy on motor function in 12 literatures were compared, and the overall OR value was 1.34 (1.00, 1.69); the results of the heterogeneity

test were $Q = 0.68$, $I^2 = 0.00\%$, and $P = 1.00$, indicating that there was no heterogeneity among the study groups. Among them, the lowest OR value was 1.03 (-1.02, 3.08), and the highest OR value was 1.57 (0.39, 2.74). The results of this meta-analysis demonstrated that electroacupuncture was more effective in improving motor function after SCI than conventional treatment.

In order to further observe the effect of the treatment, the heterogeneity test and the risk of bias assessment were carried out for the treatment. As can be seen from Figures 9 and 10, the heterogeneity gap among the articles was small and the accuracy was high and there was a certain gap between the two treatment effects. All literatures have been tested by Western blotting, and conventional treatment of SCI had not shown a good therapeutic effect on the improvement of motor function after SCI in rats. Figure 11 shows the funnel plot. The funnel asymmetric linear

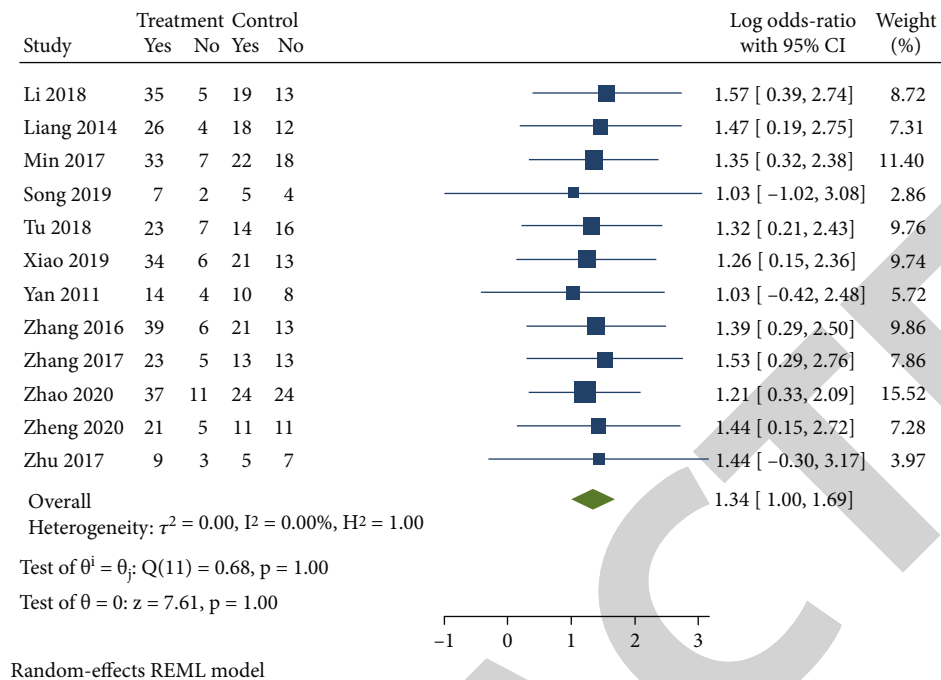


FIGURE 8: Forest plot of individual studies and pooled studies on motor function. CI: confidence interval; df: degree of freedom.

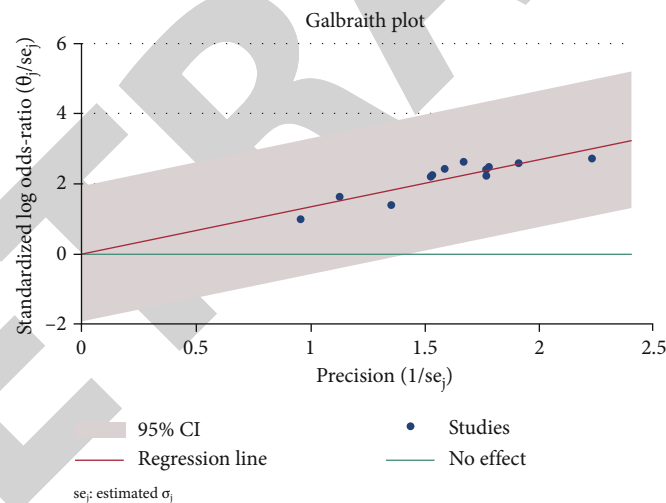


FIGURE 9: Galbraith's test chart of motor function treatment effect.

regression analysis can further observe that there was no risk of bias in the study, and each study was relatively concentrated, which proved that the results had high accuracy.

3.5. Influence of Pathway Function. In the 12 included literatures, the comparison results of conventional treatment and electroacupuncture in the treatment of SCI in the experiment were analyzed. Figure 12 is a forest plot of individual studies and pooled studies for two treatments. Comparing the effects of electroacupuncture and conventional therapy on signaling pathways in 12 literatures, the overall OR value was 1.24 (0.91, 1.57). The results of the heterogeneity test

were $Q = 3.62$, $I^2 = 0.00\%$, and $P = 0.98$, indicating that there was no heterogeneity among the study groups. Among them, the lowest OR value was 0.69 (-0.96, 2.34), and the highest OR value was 1.89 (0.69, 3.10). The results of this meta-analysis demonstrated that electroacupuncture had a better effect on the improvement of signaling pathways after SCI than conventional treatment.

In order to further observe the effect of the treatment, the heterogeneity test and the risk of bias assessment were carried out for the treatment. Figures 13 and 14 are the heterogeneity test graphs. It can be observed that the heterogeneity gap between the studies was small and the accuracy

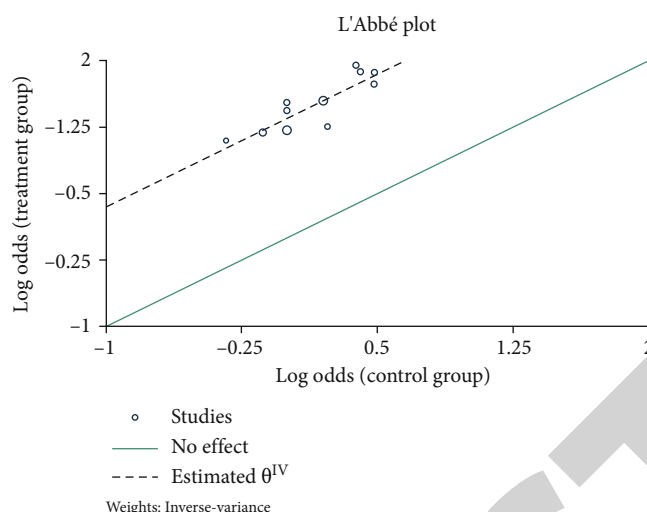


FIGURE 10: Heterogeneity test chart of motor function treatment effect.

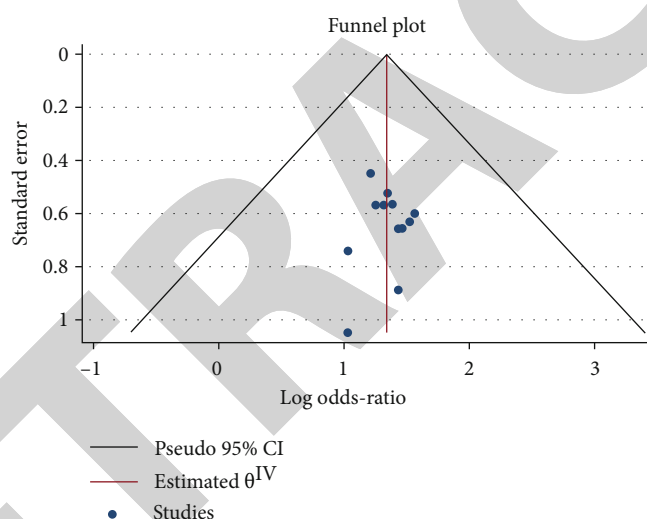


FIGURE 11: Funnel plot of motor function treatment effect.

was high, showing that there was a certain gap between the two treatment effects. All literatures had been tested by Western blotting, and the effect of conventional SCI treatment on the improvement of signaling pathways after SCI in rats was worse than that of electroacupuncture. Figure 15 shows the funnel plot. The funnel asymmetric linear regression analysis can further observe that there was no risk of bias in the study, and each study was relatively concentrated, which proved that the results had high accuracy.

3.6. Reliability Analysis. Sensitivity analysis was performed by changing the analysis model. The results of meta-analysis showed that there was no significant change in the summary results of different analysis models. It showed that the results of the inclusion in the literature had good stability. Model analysis such as funnel asymmetric linear regression analysis can also show that the consistency of the study's risk of bias verification was good.

4. Discussion

EA is a treatment method based on traditional acupuncture at Jiaji point, supplemented by pulsed current stimulation. It not only has the function of the traditional "acupuncture" technique but also can stimulate the muscles, blood vessels, nerves, and biological currents in the human body, thus forming the "field effect" of the current. It has a dual effect and is effective for many clinical diseases, such as SCI-related diseases, cervical and lumbar spine diseases, diabetes, and urinary and reproductive system diseases [14]. The therapeutic effect of EA at Jiaji point on SCI can be viewed from both Chinese and Western medicine perspectives. From the perspective of traditional Chinese medicine, since Jiaji point is adjacent to Beishu point and the two meridians of the governor meridian and the Zutaiyang bladder muscle meridian overlap here, so it is located in the transformation center of Qi in channels and organs. After acupuncture, it can be

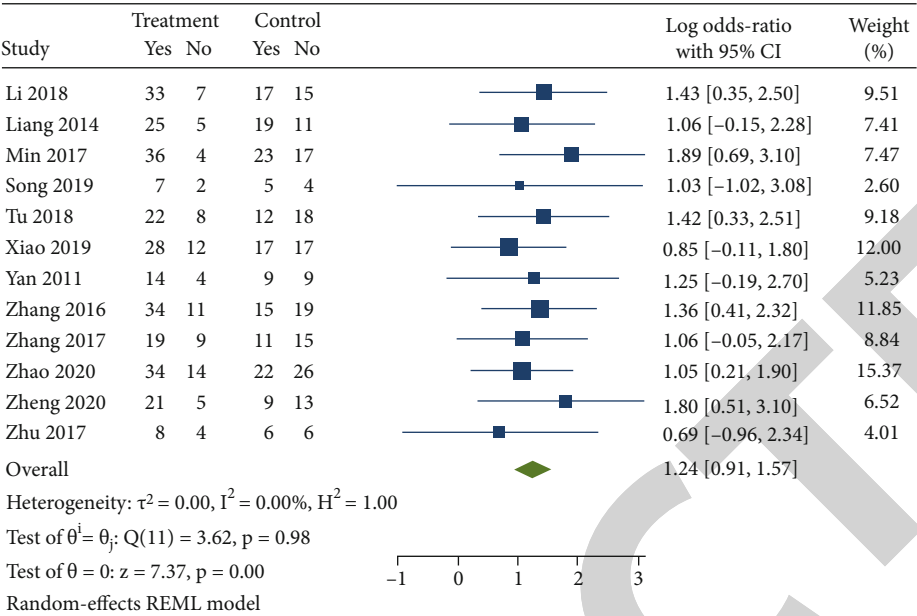


FIGURE 12: Forest plot of individual studies and aggregated studies of signaling pathways. CI: confidence interval; df: degree of freedom.

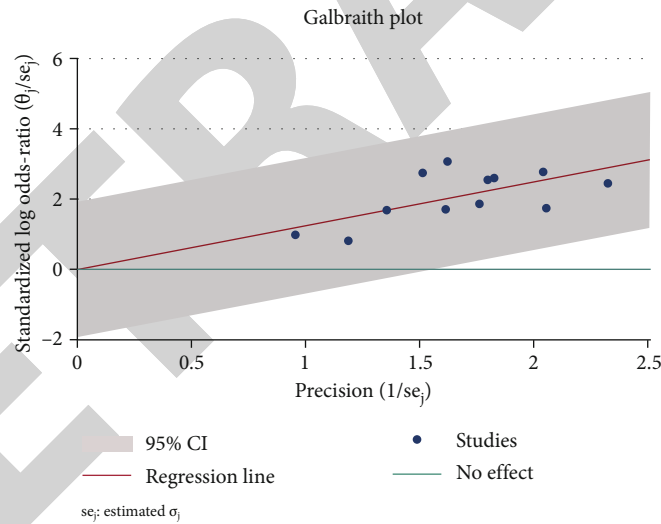


FIGURE 13: Galbraith's test diagram of the improvement effect of the signaling pathway.

adjusted to smooth the Qi and rectify the Yang, so as to harmonize the viscera and smooth the Qi and blood, thereby treating the SCI. After intervention of EA, it can expand the scope of action of acupuncture to enhance the therapeutic effect [15]. From the perspective of Western medicine, EA at Jiaji point can improve the internal and external microenvironment of neurons in the SCI area, inhibit their inflammatory response, and promote the secretion of neurotrophic protective factors. It can effectively inhibit the synthesis of prostaglandins and vascular endothelin, improve the blood oxygen deficiency in the internal environment, and protect the spinal cord barrier. In addition, it can balance the disorder of calcium ions and reduce the damage

of excitatory amino acids to synapses. The changes of these cascade reactions can effectively alleviate the secondary injury of SCI, thereby promoting the regeneration of damaged myelin sheath and neural functional recovery. A large number of studies have shown that Jiaji point is the capsule where the posterior branch of spinal nerves runs [16, 17]. After SCI, the axons are damaged and the neural circuits are destroyed, because the axon regeneration capacity of the central nervous system is limited. Therefore, sprouting injured axons and their residual collaterals is a key step in the process of neuronal polarization and a prerequisite for repairing spinal nerve function, during which RhoA plays an important role. RhoA, a member of the small GTPase

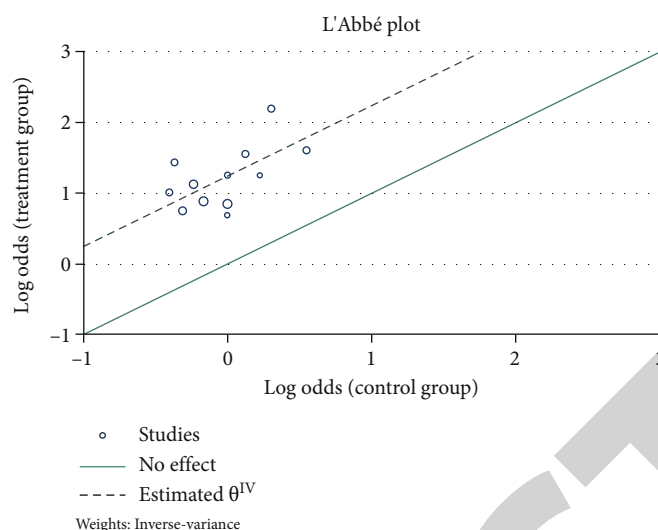


FIGURE 14: Heterogeneity test diagram of the improvement effect of signaling pathway.

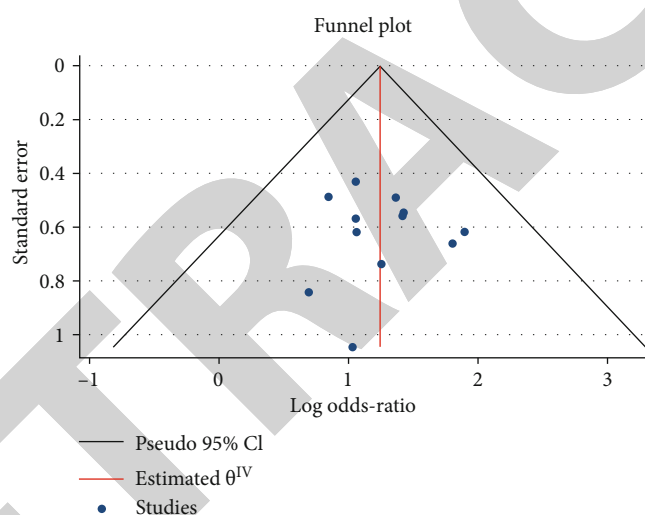


FIGURE 15: Funnel plot of signal pathway improvement effect.

family, participates in a variety of cellular regulatory functions, controls cell morphology, and is closely related to cell motility, transcriptional regulation, growth and development, and its cytoskeleton rearrangement. The morphology of a cell depends on the activity of its cytoskeleton, and microtubules, neurofilaments, and actin filaments are important components of spinal cord axons. Activated RhoA induces the accumulation of actin networks in damaged axons, thereby regulating their cytoskeleton. In addition, it is a negative regulator of damaged axon growth, affecting neuronal polarization and the initiation of surviving axons. Relevant meta-study analysis shows that RhoA inhibitors can reduce the formation of syringomyelia in the spinal cord. For example, LINGO-I-FC treatment can promote functional recovery after corticospinal tract transection, and BA-210 can increase the number of damaged areas and density of nerve fibers, thereby promoting axonal sprouting and significantly improving motor function.

Therefore, inhibiting the expression of RhoA may be one of the important targets for restoring SCI. In previous related experimental studies, it was found that EA at Jiaji point could regulate the expression of NgR in the spinal cord transection area of SCI rats and promote the secretion of myelin basic protein, so as to achieve the purpose of repairing damaged myelin. As a small molecule protein of its related downstream pathways, RhoA is closely related to the inhibitory microenvironment after SCI, and the changes of its protein expression are related to different cells, different time, and different treatment methods.

In this work, 12 literatures with a total of 692 rats were included through meta-analysis, and the heterogeneity test was performed on the 12 included articles. The results showed that there was no heterogeneity among the studies in terms of neurological recovery ($I^2 = 0.00\%$); in terms of motor function, there was no heterogeneity among the studies ($I^2 = 0.00\%$); and there was no heterogeneity ($I^2 = 0.00\%$).

in the functional recovery. In terms of nerve recovery, electroacupuncture was more effective than conventional treatment for nerve recovery after SCI. Comparing the effects of electroacupuncture and conventional treatment on motor function, the total OR value was 1.34 (1.00, 1.69), and electroacupuncture could better improve the recovery of motor function in rats after SCI. On the other hand, the conventional treatment of SCI did not have a good inhibitory effect on the expression of RhoA protein signal in rats, and it was found that the influence of the signal pathway was not obvious according to the change of the inhibitory effect. SCI+EA had a good inhibitory effect on the expression of RhoA protein signal in rats and had a positive effect on the signal pathway.

In conclusion, this meta-analysis had synthesized and evaluated the effects of conventional treatment and EA +SCI on neurological recovery and related pathways after SCI in rats and can provide evidence-based recommendations for clinical practice guidelines. In clinical research, conventional treatment can be combined with EA to accurately evaluate complex conditions, providing more accurate reference for subsequent treatment.

5. Conclusion

In this work, relevant literatures on the effects of conventional therapy and electroacupuncture on the recovery of nerve function and related pathways after SCI were screened and included in a meta-analysis, to explore the effectiveness of electroacupuncture for neurological recovery after SCI. Meta-analysis results confirmed that electroacupuncture was more effective than conventional therapy for spinal cord injury in rats; and electroacupuncture was more effective than conventional therapy in inhibiting pathways. However, for the potential application of electroacupuncture for neurological recovery after SCI, it was necessary to further formulate unified treatment standards. At the same time, more samples and higher-quality research were needed for further in-depth demonstration to provide a more accurate and effective basis for clinical practice.

Data Availability

All data, models, and code generated or used during the study appear in the submitted article.

Conflicts of Interest

There is no conflict of interest in this paper.

Authors' Contributions

Zengjiao Lai and Huihui Liu contributed equally to this work.

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